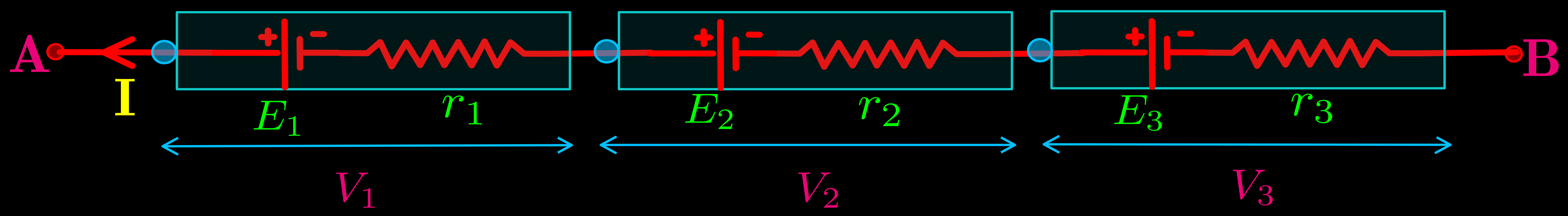
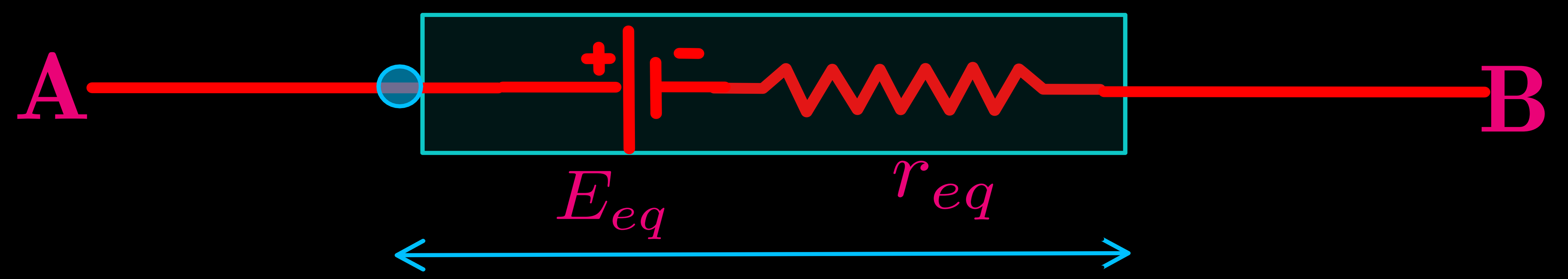
CELLS IN SERIES

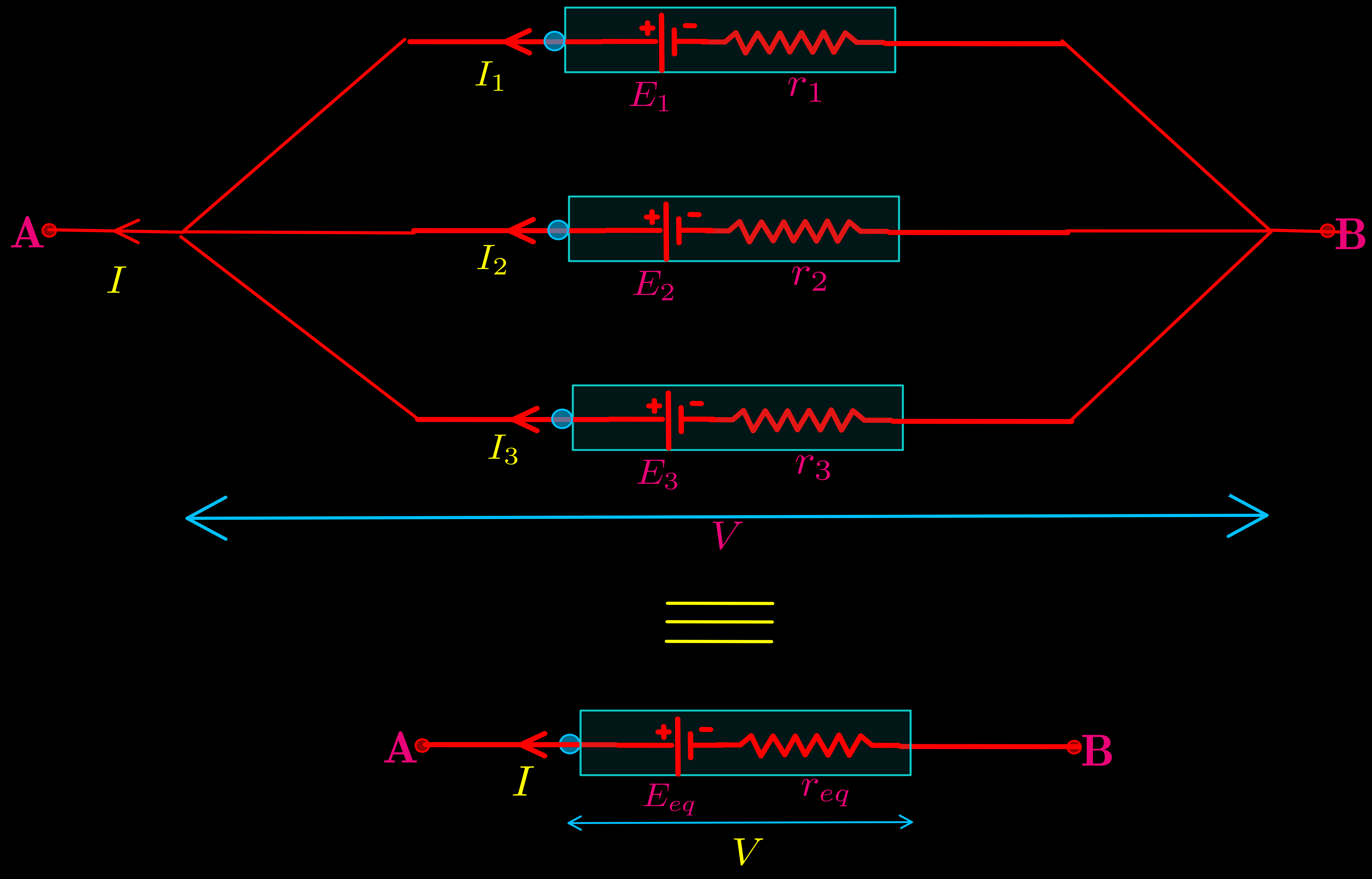


$$\equiv$$



$$V_{eq} = V_1 + V_2 + V_3$$

CELLS IN PARALLEL



KIRCHHOFF'S RULES

Chapter [3] - Current Electricity
Class - XII

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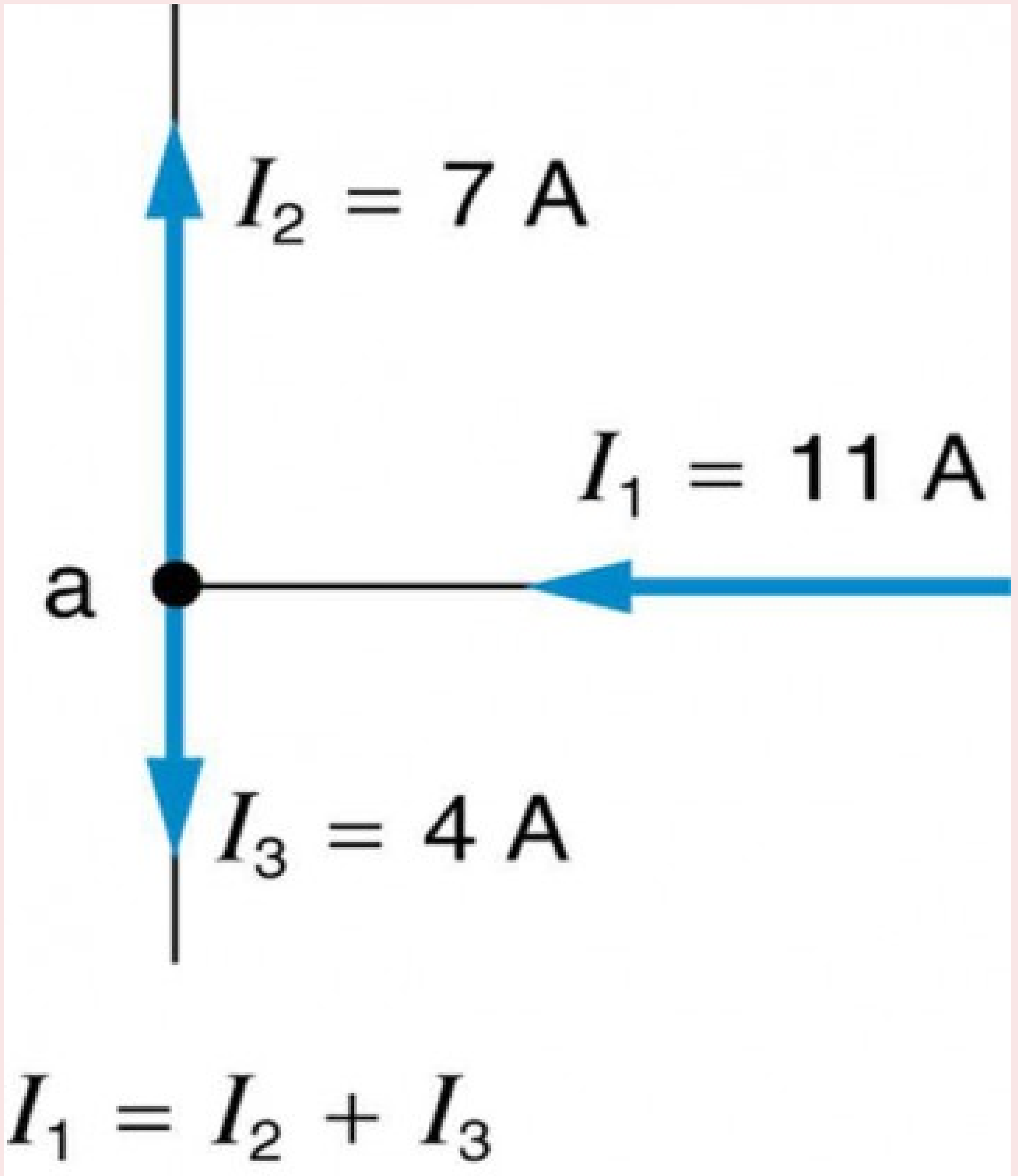
June 8, 2025

KIRCHHOFF'S RULES

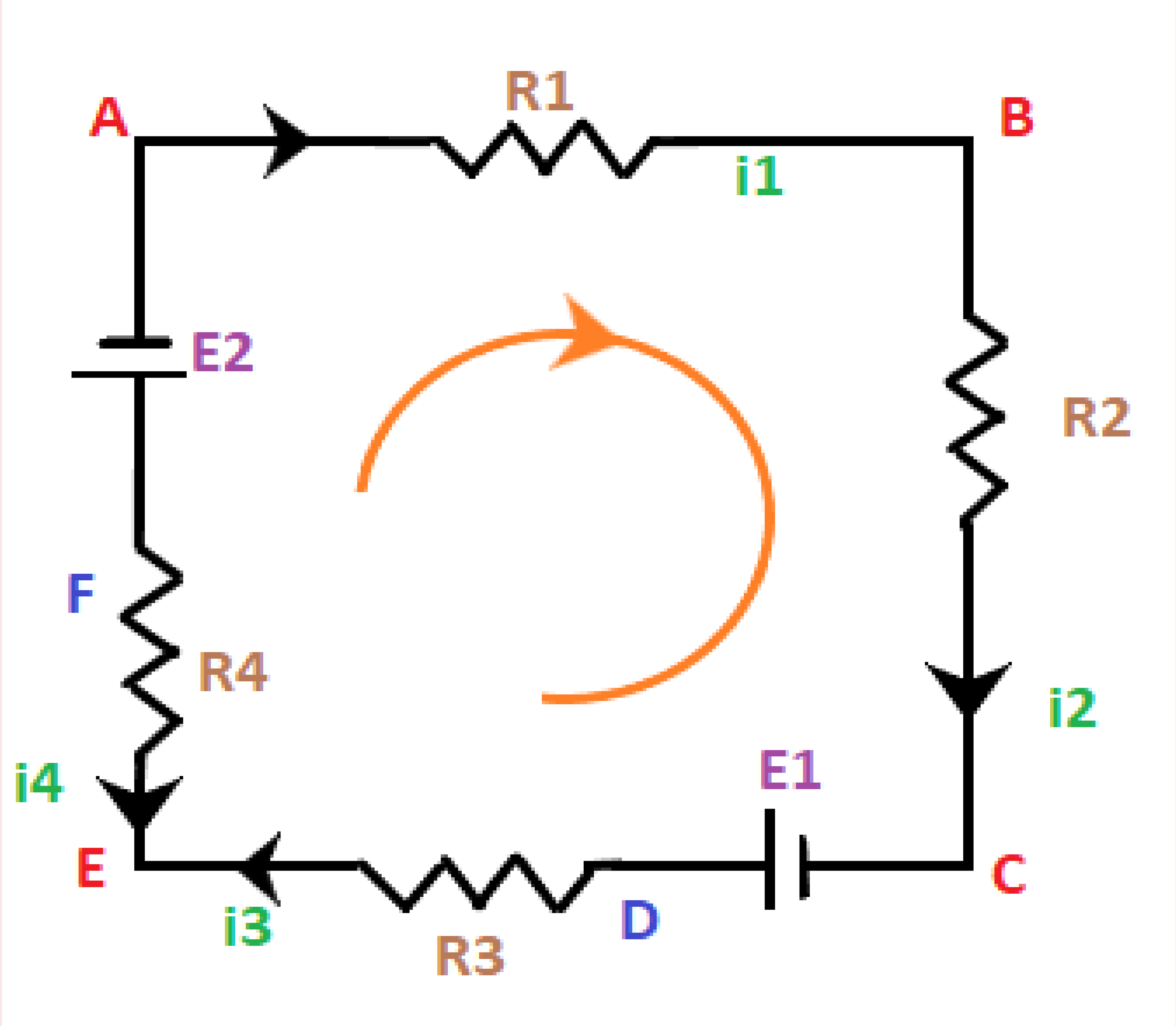


Junction rule:

- At any **junction**, the **sum of the currents entering** the junction is **equal** to the **sum of currents leaving** the junction. This is a consequence of **charge conservation**.

$$\sum I = 0$$

KIRCHHOFF'S RULES



$$E_1 - E_2 =$$
$$i_3 R_3 - i_4 R_4 + i_1 R_1 + i_2 R_2$$

Loop rule:

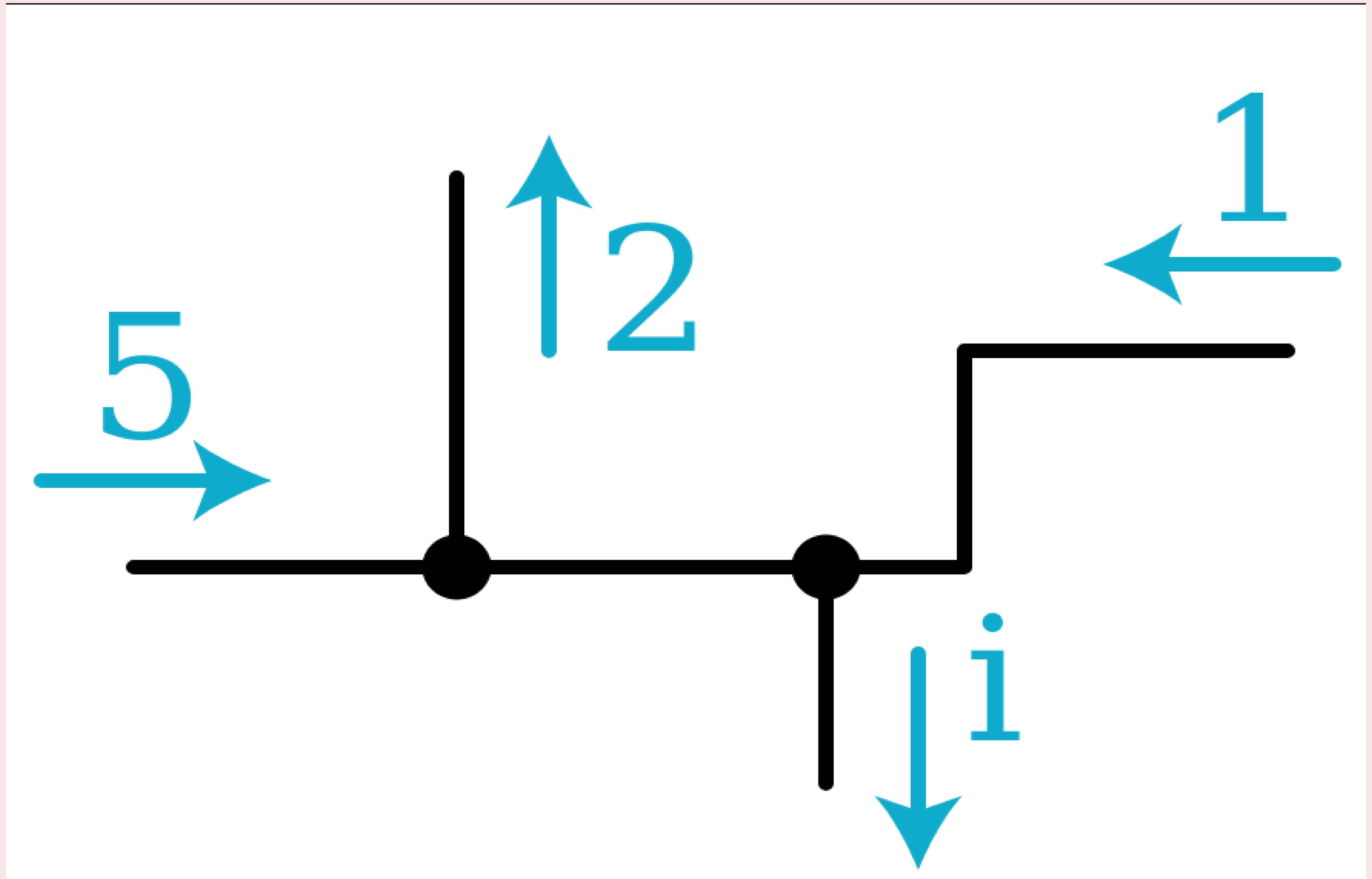
- It states that around each **loop** in an electric circuit the **sum of the emf's** is **equal** to the **sum of the potential drops, or voltages across each of the resistances**, in the same loop.

$$\sum \xi = \sum IR$$

- The **loop rule** is an application of **conservation of energy**. In a closed loop, whatever energy is supplied by emf must be transferred into other forms by devices in the loop.

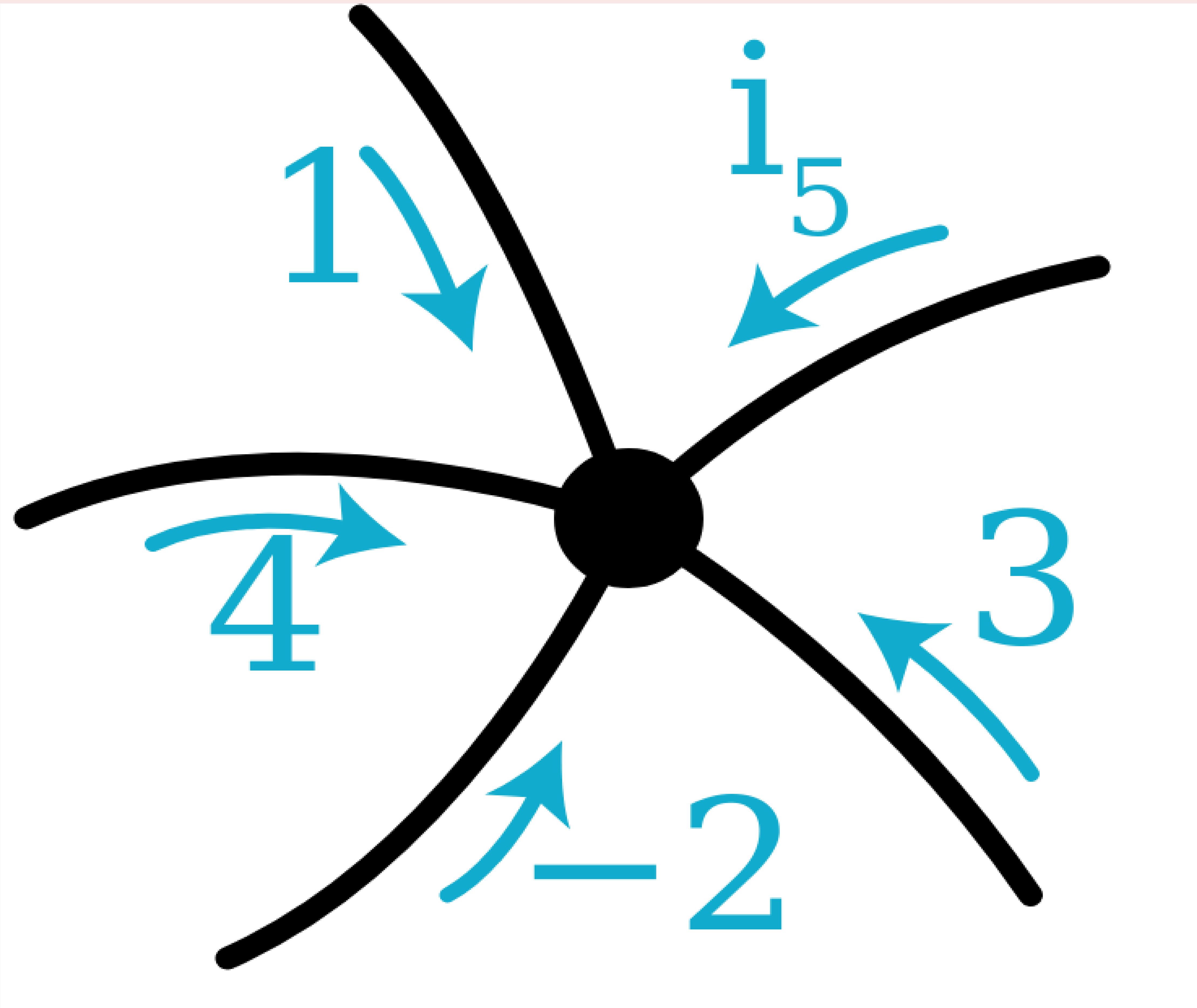
Kirchhoff's Current Law - concept checks

Problem 1: What is i?



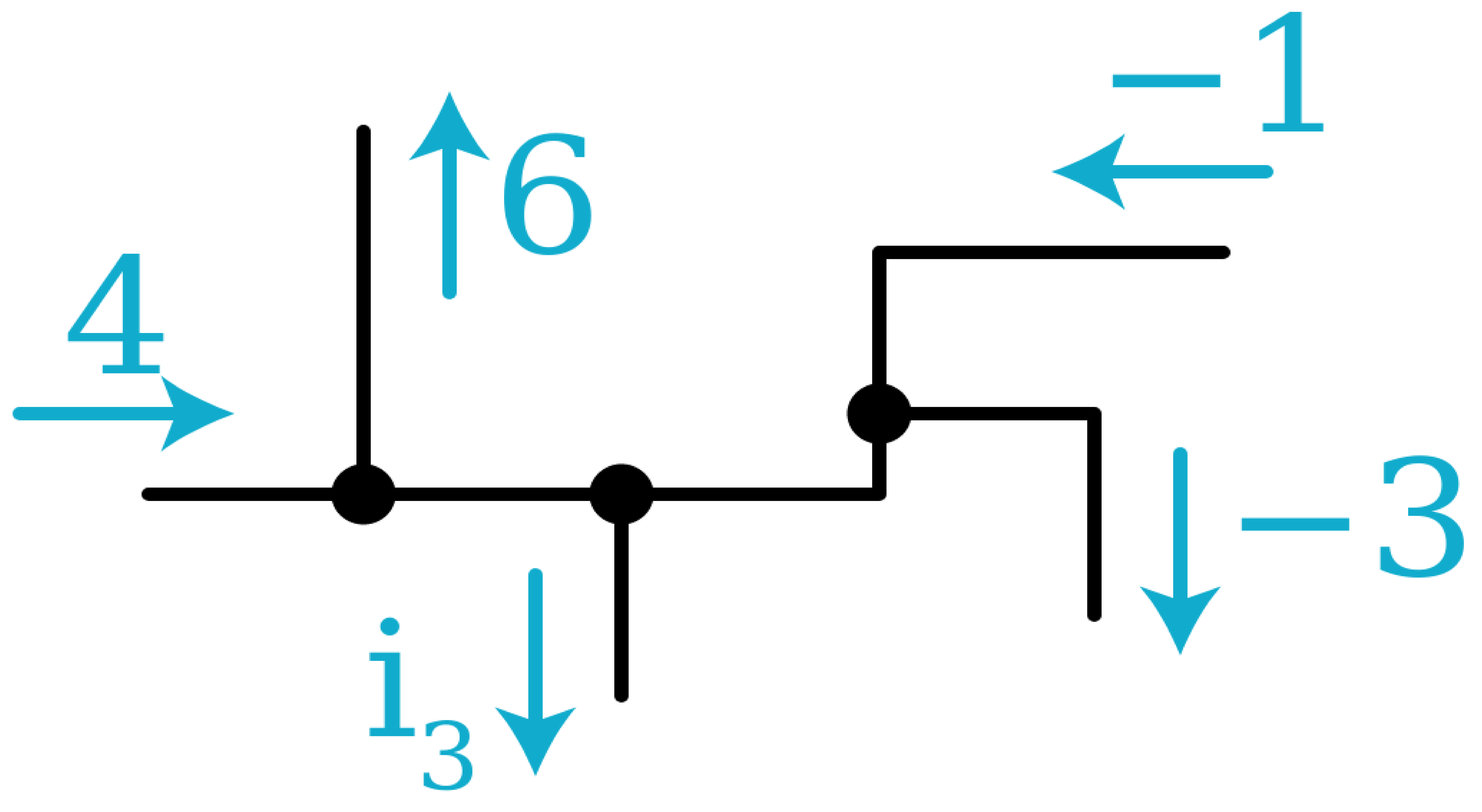
Kirchhoff's Current Law - concept checks

Problem 2 : What is i_5?



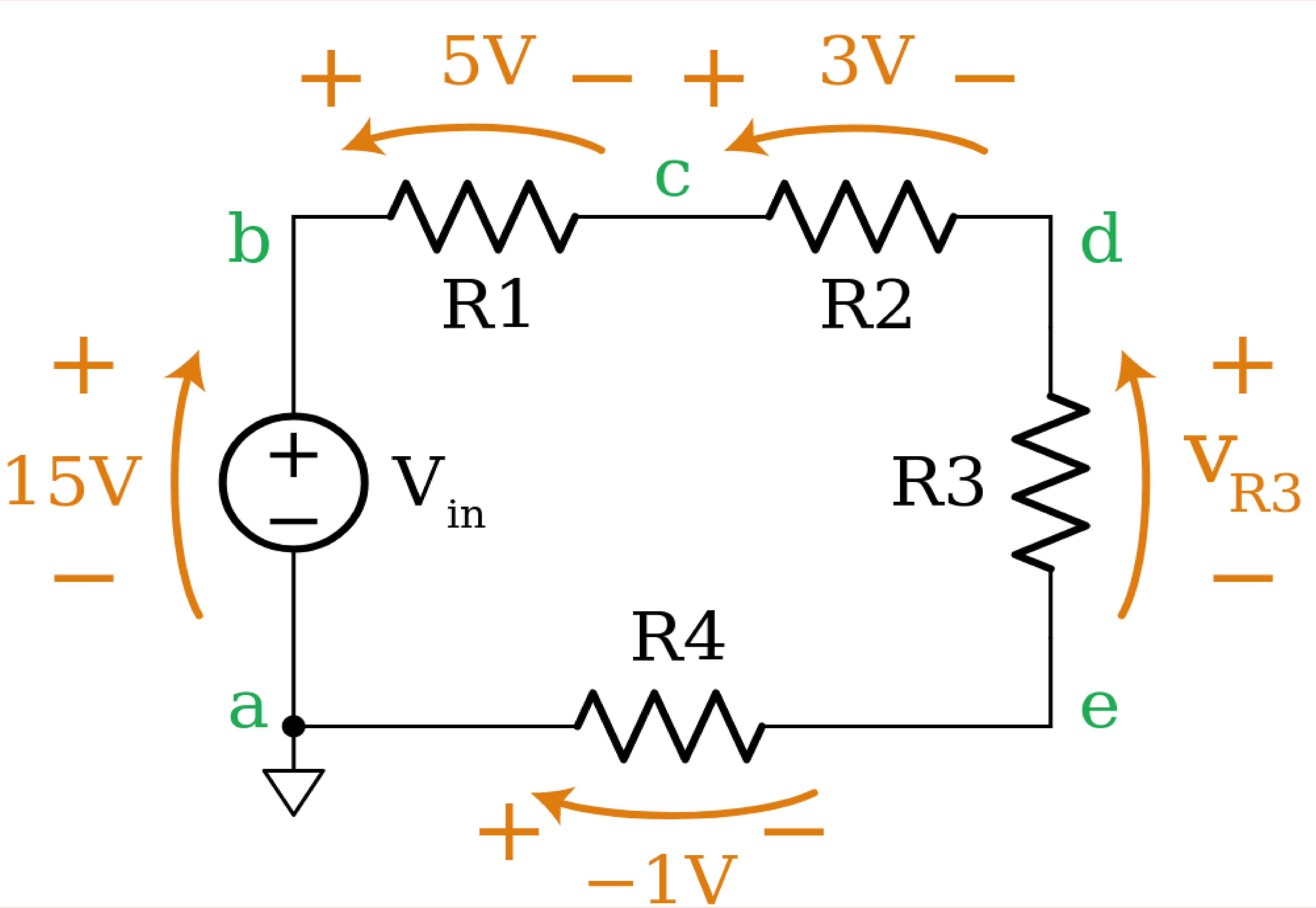
Kirchhoff's Current Law - concept checks

Problem 3: What is i_3?



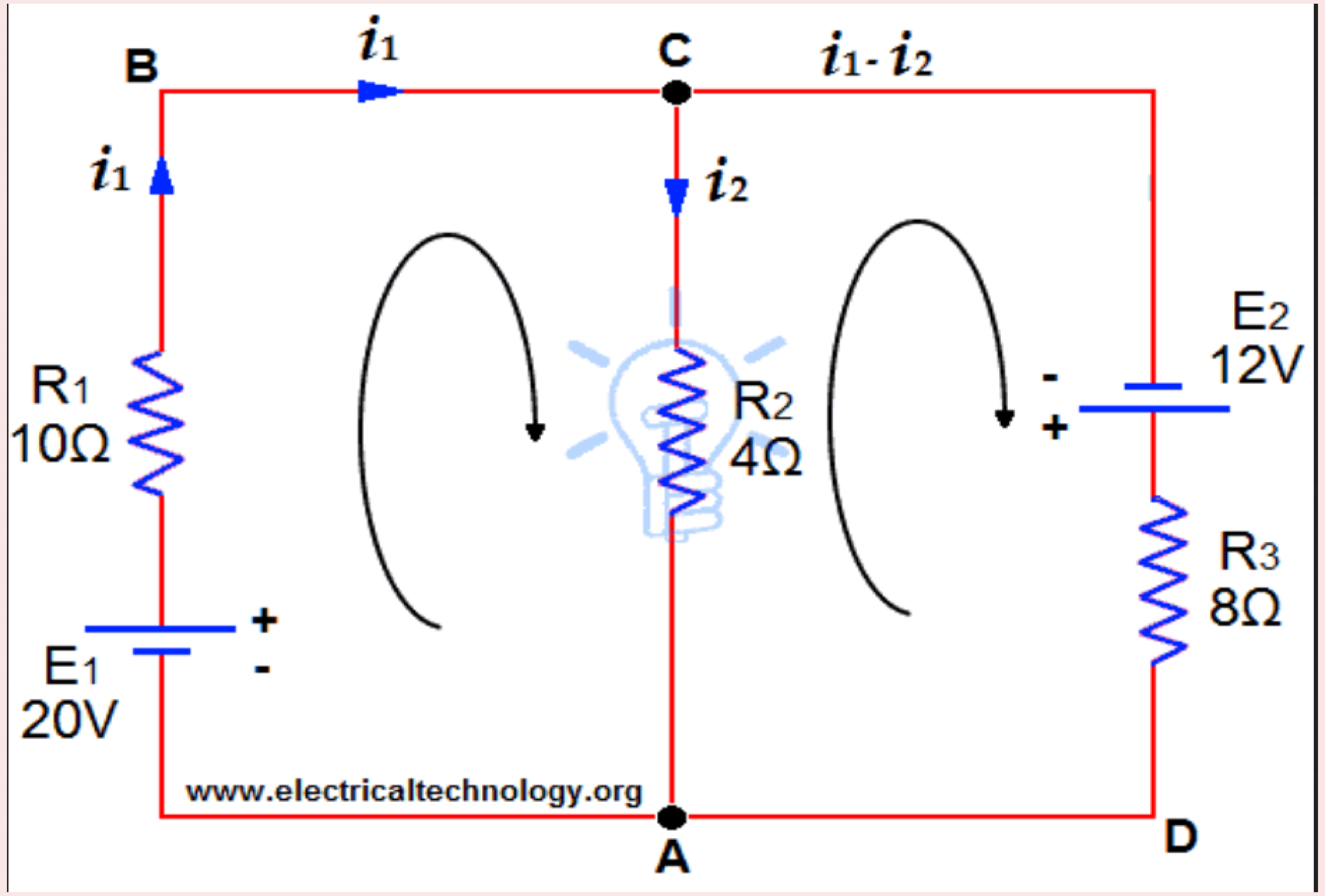
Kirchhoff's Voltage Law - concept check

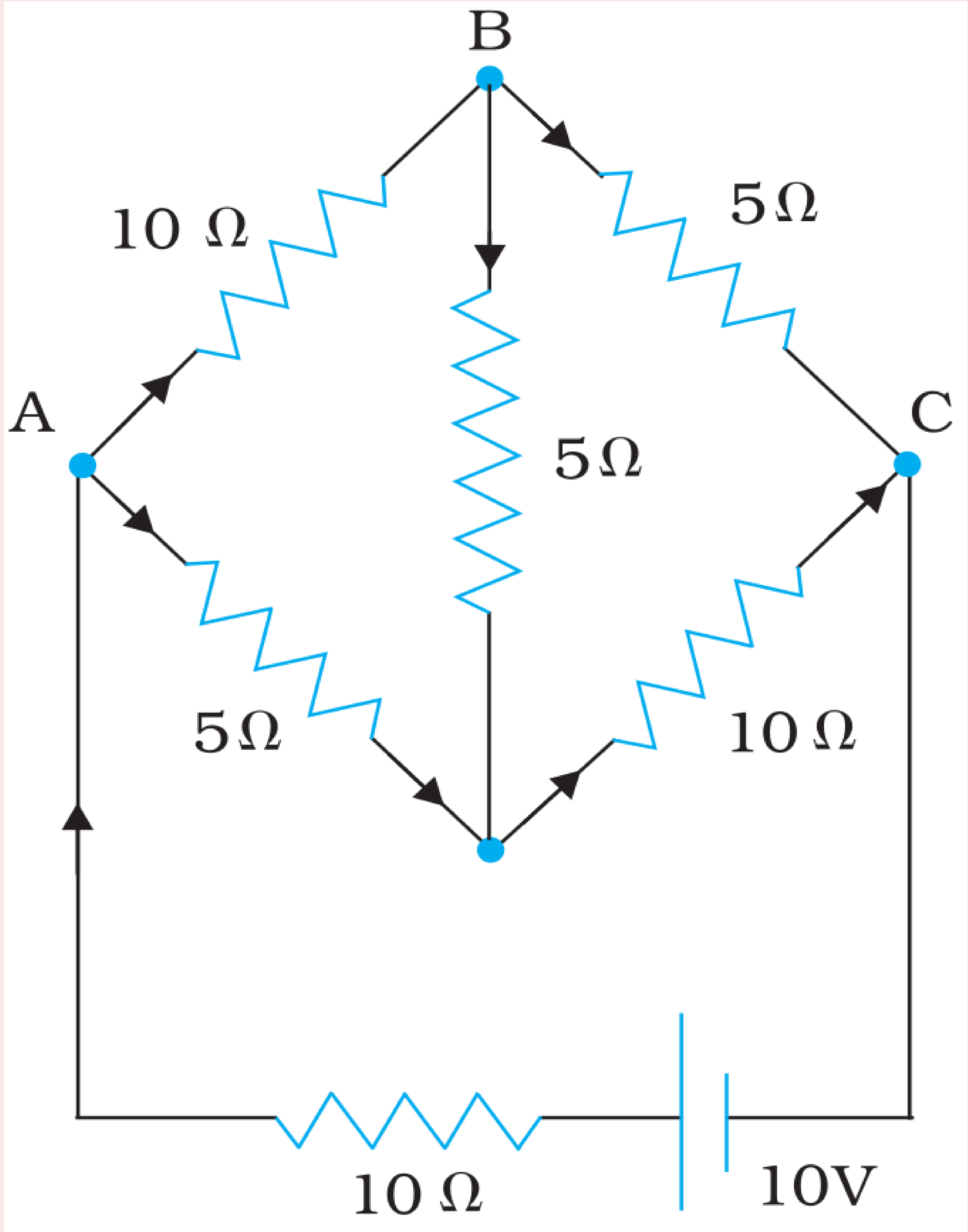
Problem 4: What is V_{R3}?



Kirchhoff's Voltage Law - concept check

Problem 5: Resistors of $R_1 = 10\Omega$, $R_2 = 4\Omega$ and $R_3 = 8\Omega$ are connected up to two batteries as shown. Find the current through each resistor.

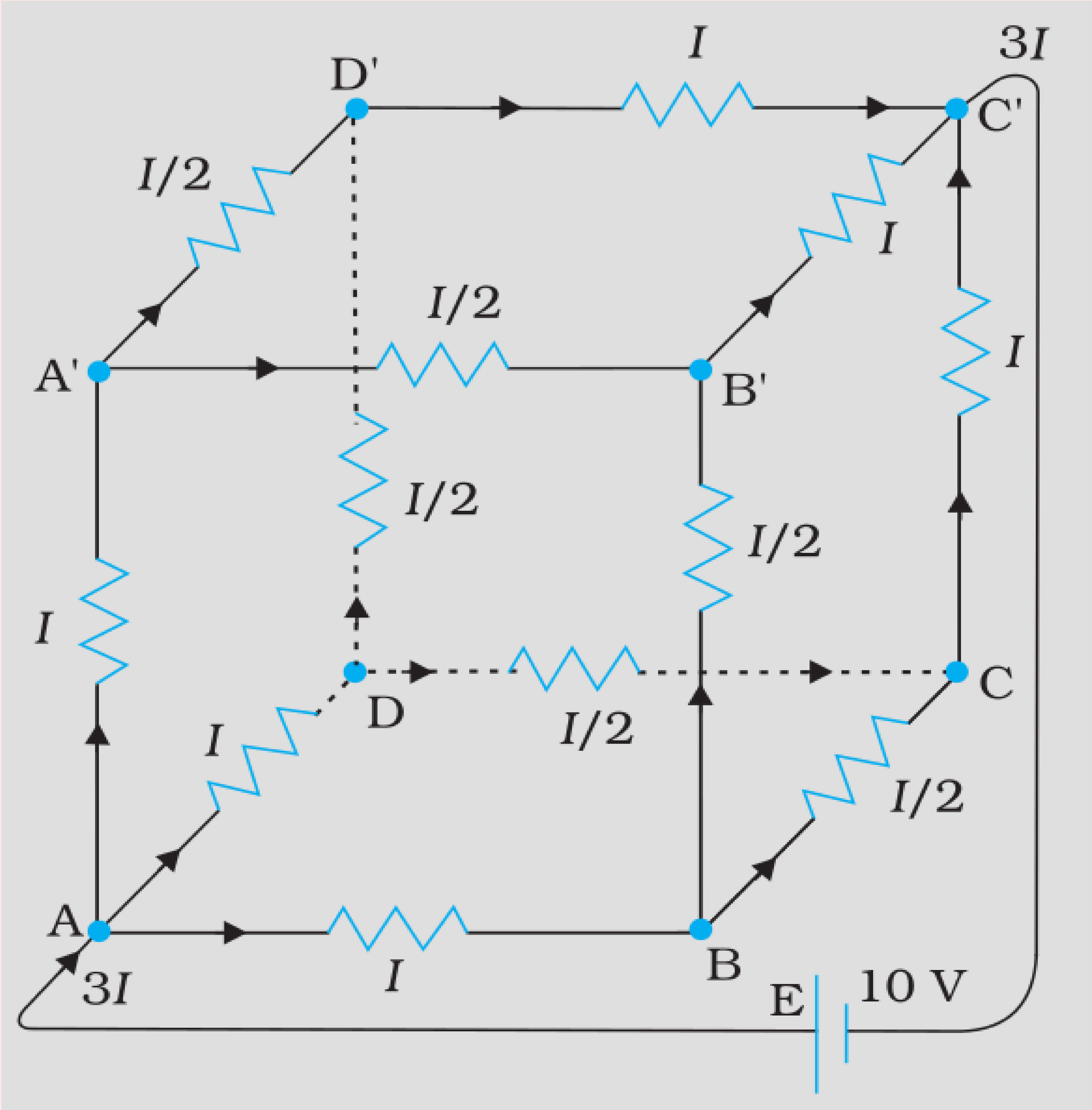




Problem 6:

Determine the current in each branch of the network shown in Fig.

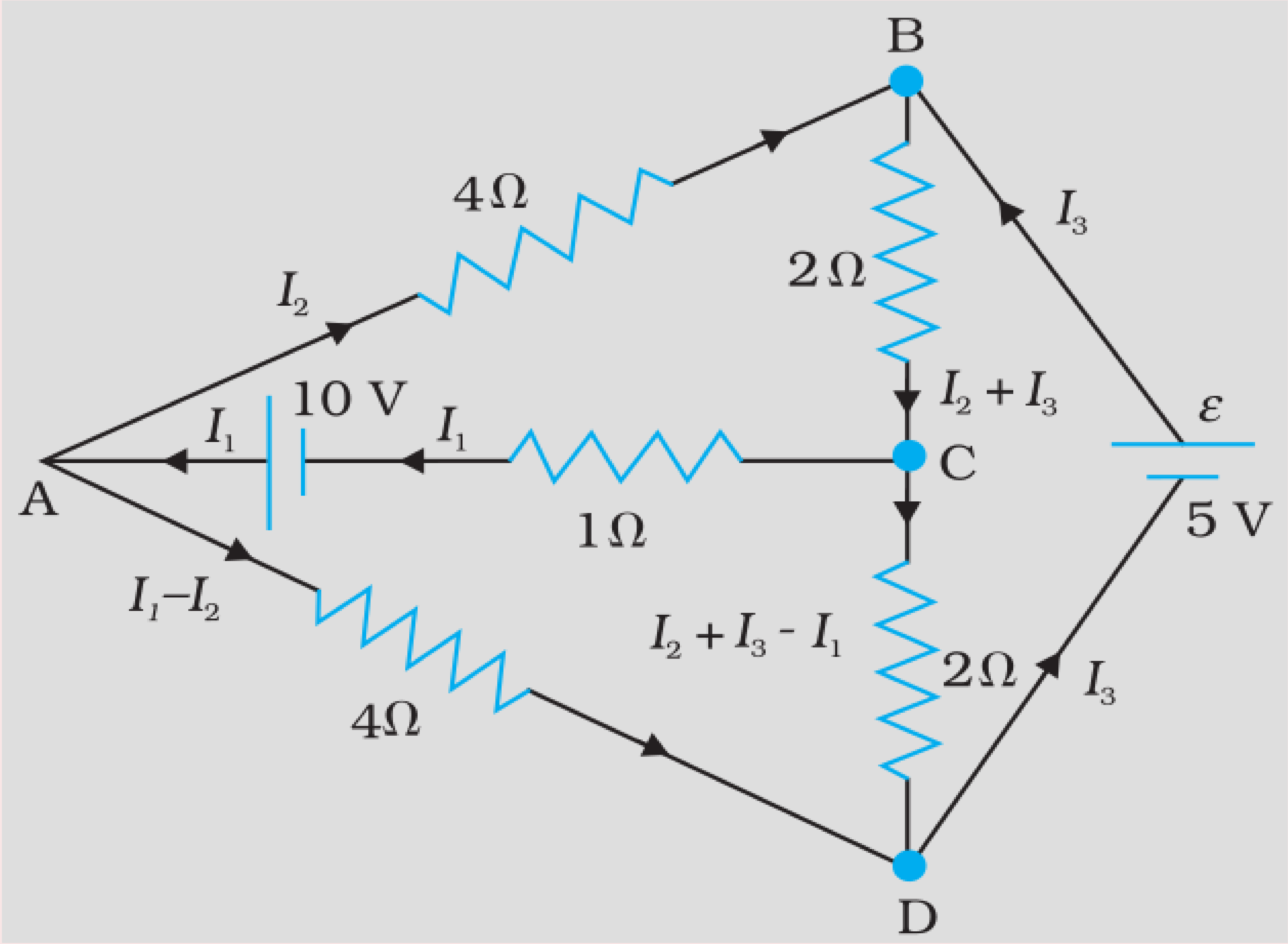
Kirchhoff's Laws



Problem 7:

A battery of 10 V and negligible internal resistance is connected across the diagonally opposite corners of a cubical network consisting of 12 resistors each of resistance 1 Ω (Fig. 3.23). Determine the equivalent resistance of the network and the current along each edge of the cube.

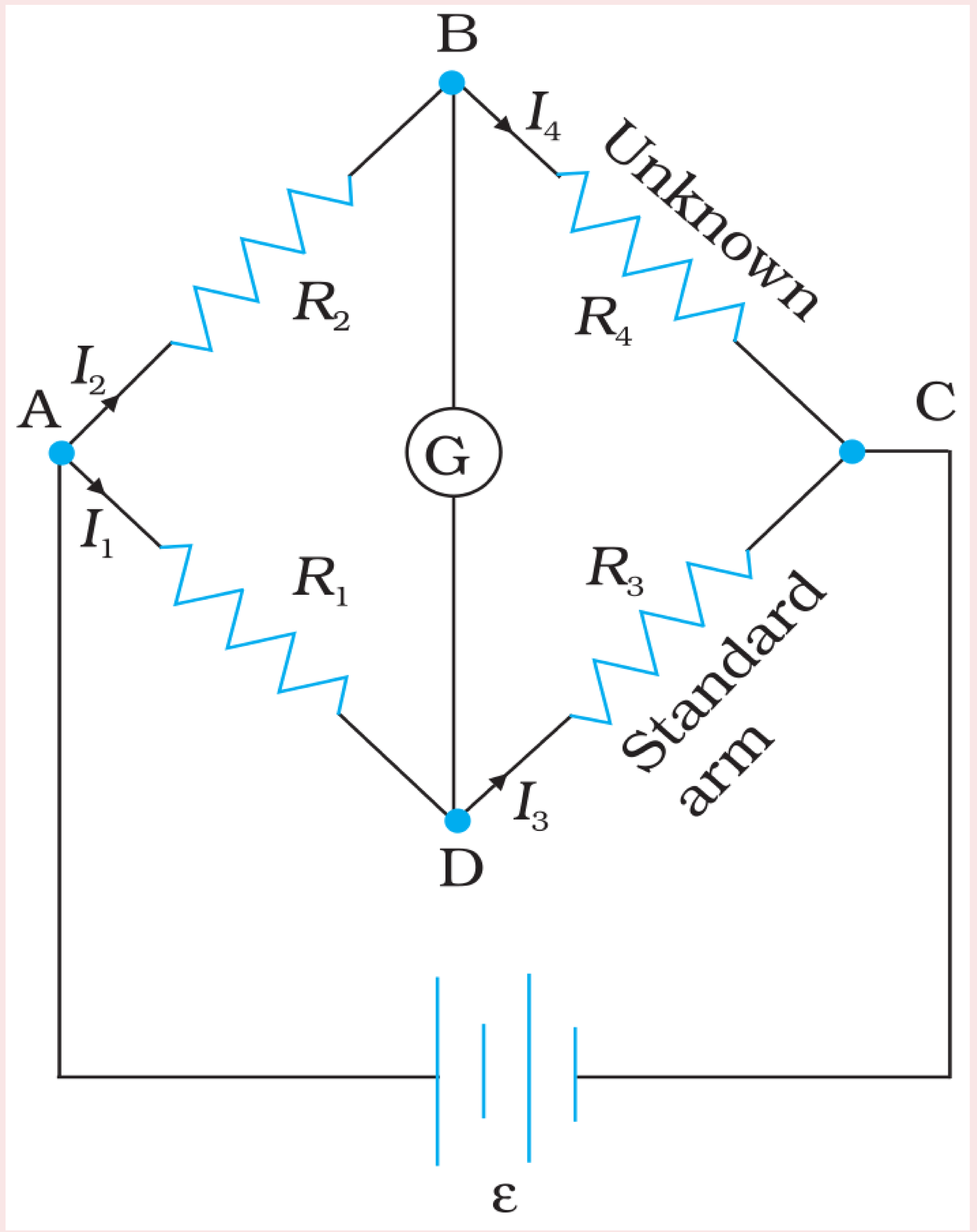
Kirchhoff's Laws



Problem 8:

Determine the current in each branch of the network shown in Fig.

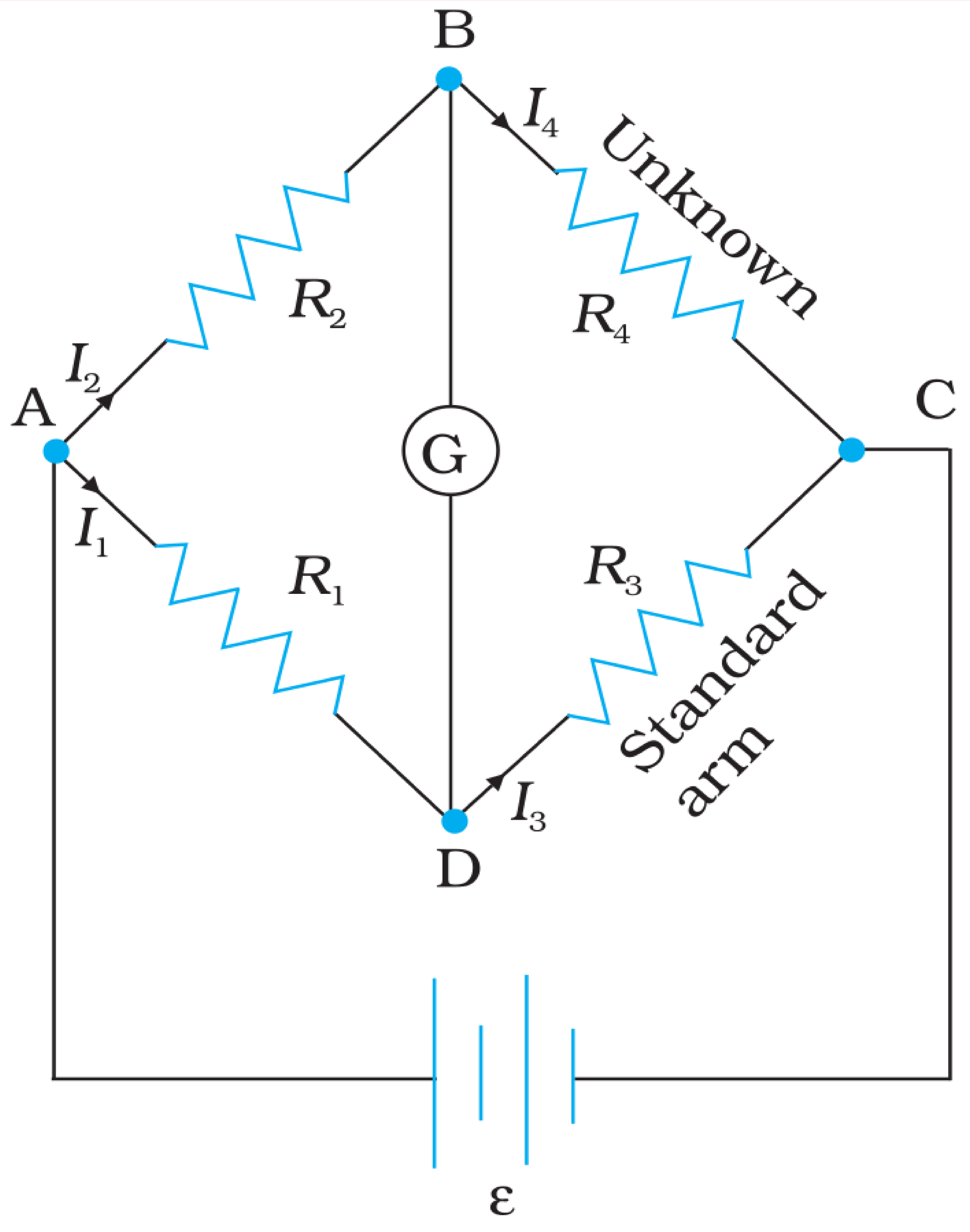
WHEATSTONE BRIDGE



Construction of Wheatstone Bridge:

- The **Wheatstone bridge has four resistors** R_1, R_2 R_3 and R_4.
- Across one pair of diagonally opposite points (**A and C** in the figure) a **source (battery) is connected.**
- Between the other two vertices, **B and D**, a **galvanometer G** (which is **a device to detect currents**) is **connected**.

WHEATSTONE BRIDGE

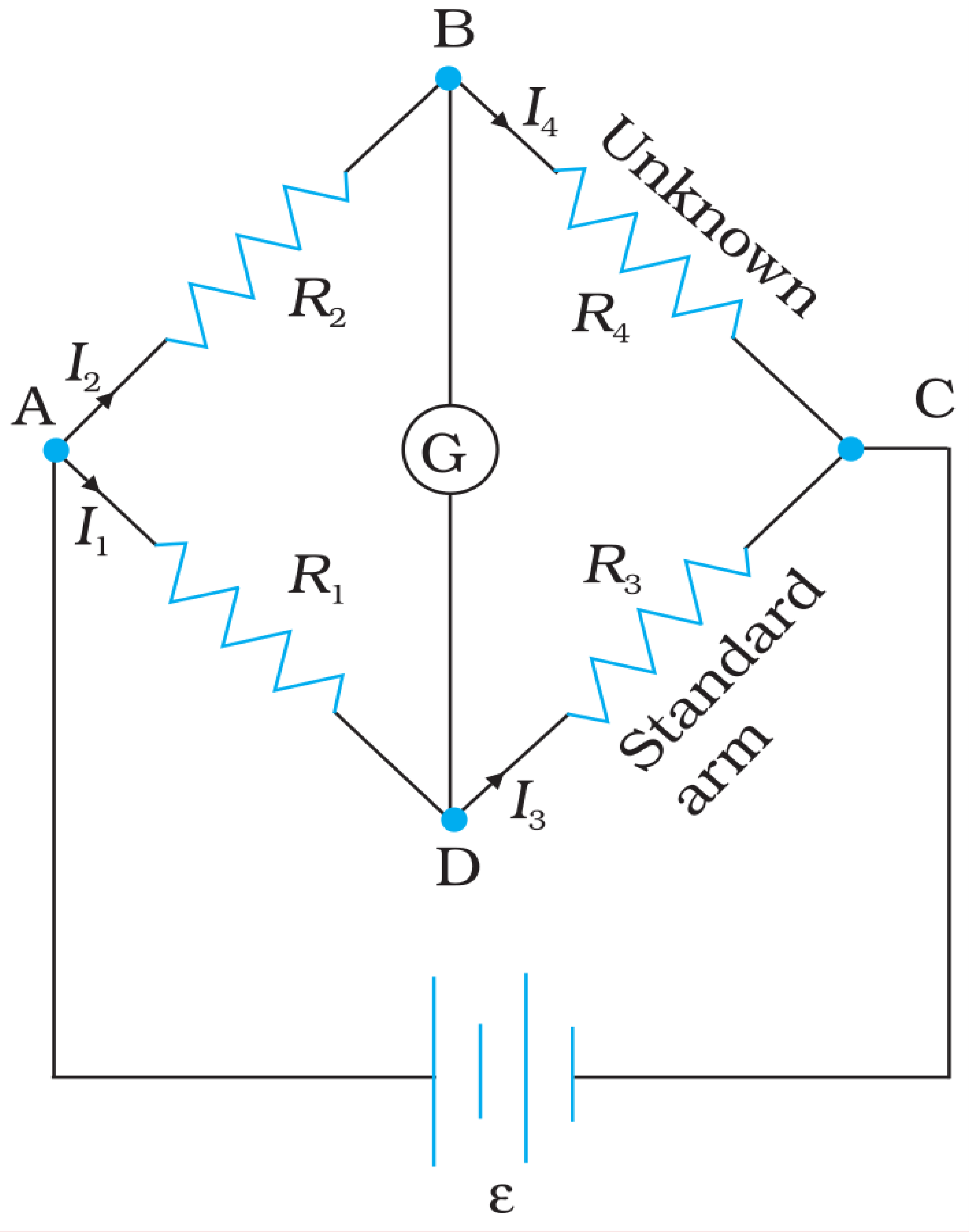


Derivaiton: Balanced Bridge

- In general there will be currents flowing across all the resistors as well as a current I_g through G.
- In the case of a **balanced bridge** the resistors are such that there is **no current through G** (or $Ig = 0$).
- In this case **using Kirchhoff's junction** rule at the junction B and D gives:

$$I_2 = I_4 \text{ and } I_1 = I_3$$

WHEATSTONE BRIDGE



- Using Kirchhoff's **loop rule** to closed **loop ABDA**

$$I_2R_2 - 0 + I_1R_1 = 0 \quad \because (I_g = 0)$$
$$\boxed{I_2R_2 = I_1R_1} \quad .....(1)$$

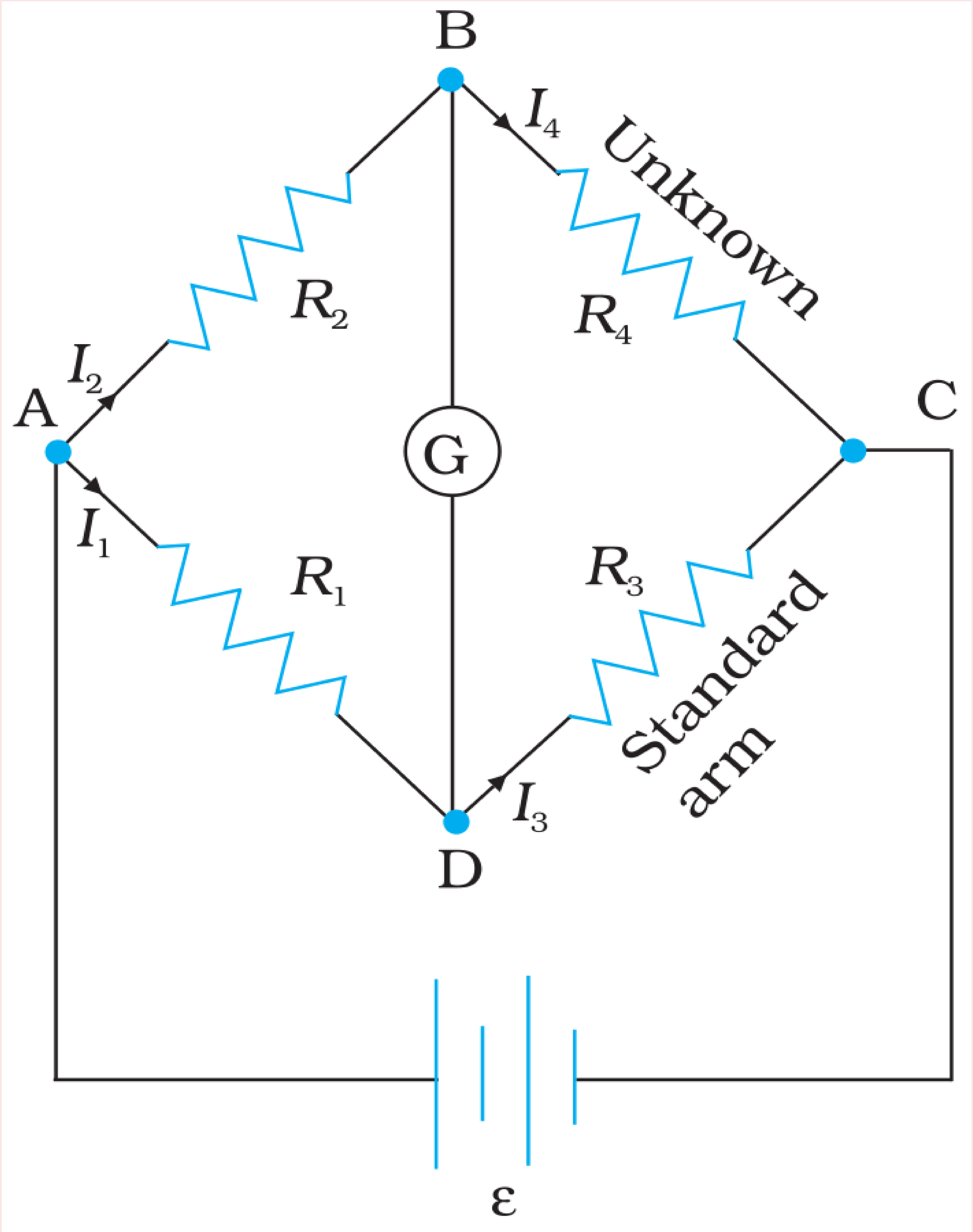
- Using Kirchhoff's **loop rule** to closed **loop BCDB**

$$I_4R_4 - I_3R_3 + 0 = 0 \quad \because (I_g = 0)$$
$$\implies I_2R_4 - I_1R_3 = 0 \quad \because (I_4 = I_2 \text{ and } I_3 = I_1)$$
$$\boxed{I_2R_4 = I_1R_3} \quad .....(2)$$

- **Dividing eq(1) by eq(2)** we get:

$$\boxed{\frac{\boldsymbol{R_2}}{\boldsymbol{R_4}} = \frac{\boldsymbol{R_1}}{\boldsymbol{R_3}}} \quad \textbf{.......(3)}$$

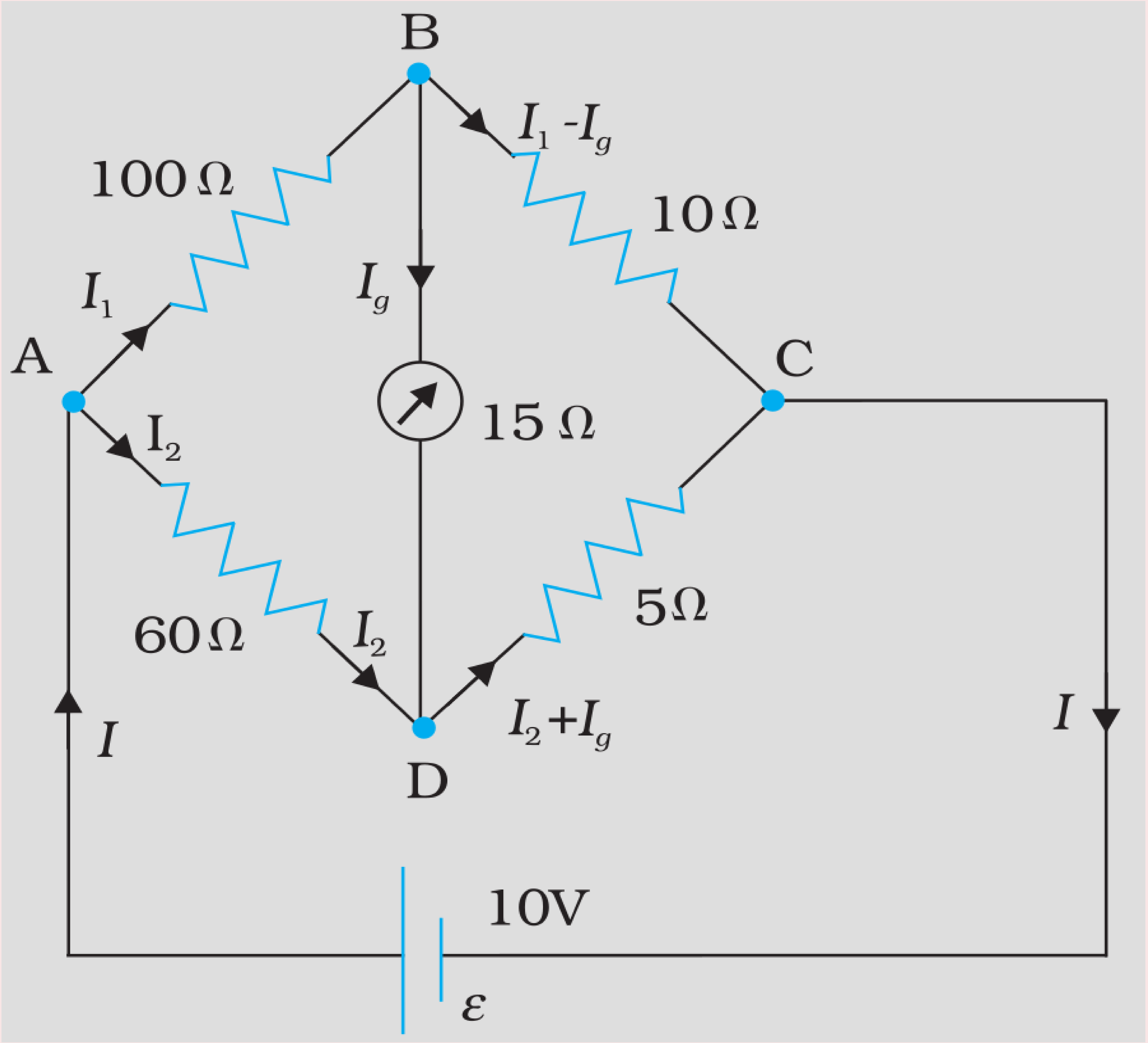
USES OF WHEATSTONE BRIDGE



- The **Wheatstone bridge** and **its balance condition** provide a practical method for **determination** of an **unknown resistance**.
- Let, $\boldsymbol{R_4}$ is the **unknown resistor**. and $\boldsymbol{R_1}$, $\boldsymbol{R_2}$ are **known resistors**.
- We go **on varying** $\boldsymbol{R_3}$ till the **galvanometer** shows a **null deflection**. The bridge then is **balanced** and the **value of the unknown resistance** $\boldsymbol{R_4}$ is given by

$$\boxed{R_4 = \frac{R_2 R_3}{R_1}} \quad ....(4)$$

WHEATSTONE BRIDGE



Example 3.8

The four arms of a Wheatstone bridge have the following resistances:
AB = 100 Ω, BC = 10 Ω, CD = 5 Ω, and DA = 60 Ω.